



WAN Technologies



Foreword

- As economic globalization and digital transformation accelerate, enterprises keep expanding their scales. More and more branches locate in different regions, with each branch network being considered as a local area network (LAN). The headquarters and branches need to cross geographical locations to communicate with each other. To better carry out services, an enterprise needs to connect these geographically dispersed branches through a wide area network (WAN).
- The development of the WAN technologies is accompanied by the continuously increased bandwidth. In the early stage, X.25 provided only the bandwidth of 64 kbit/s. Later, the digital data network (DDN) and Frame Relay (FR) increased the bandwidth to 2 Mbit/s. Synchronous digital hierarchy (SDH) and asynchronous transfer mode (ATM) further increased the bandwidth to 10 Gbit/s. Now, the current IP-based WANs provide 10 Gbit/s or even higher bandwidth.
- This course describes the development history of WAN technologies, especially the implementations and configurations of Point-to-Point Protocol (PPP) and Point-to-Point Protocol over Ethernet (PPPoE).



Objectives

- On completion of this course, you will be able to:
 - Understand the basic concepts and development history of WANs.
 - Understand PPP and PPPoE implementations.
 - Master basic PPP and PPPoE configurations.
 - Understand basic MPLS/SR concepts.



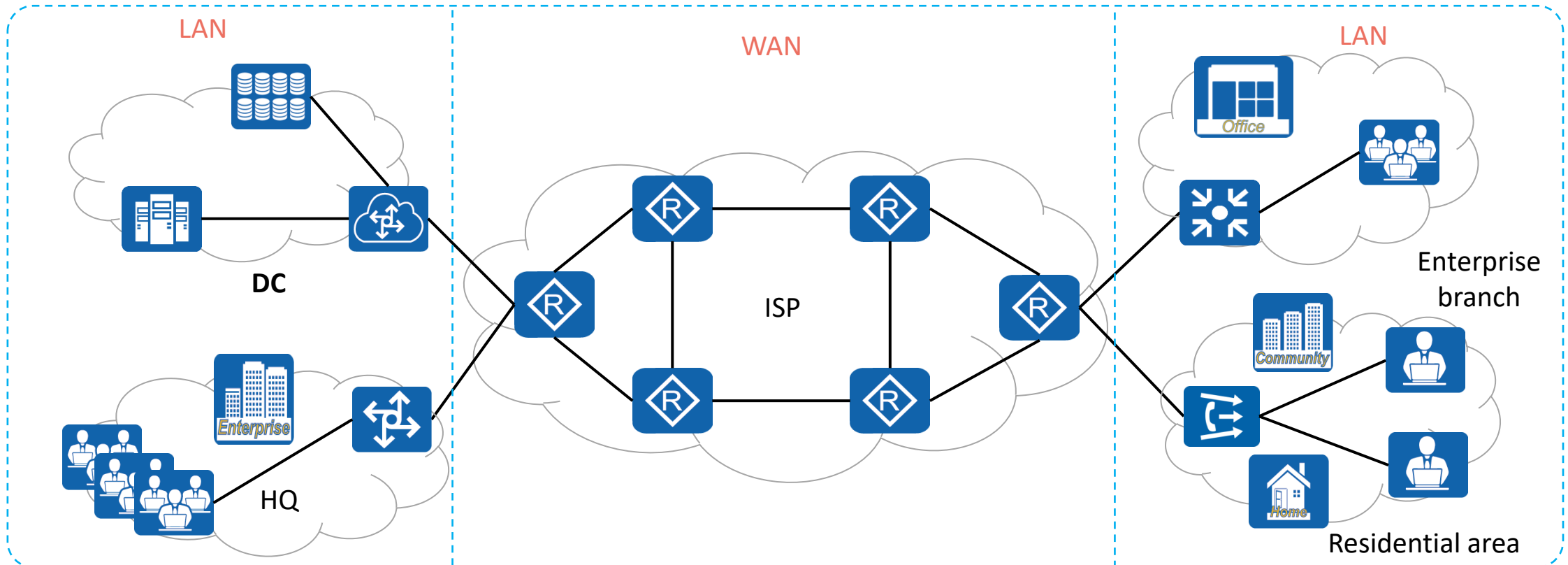
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- 1. Overview of Early WAN Technologies**
2. PPP Implementation and Configuration
3. PPPoE Implementation and Configuration
4. Development of WAN Technologies



What Is a WAN?

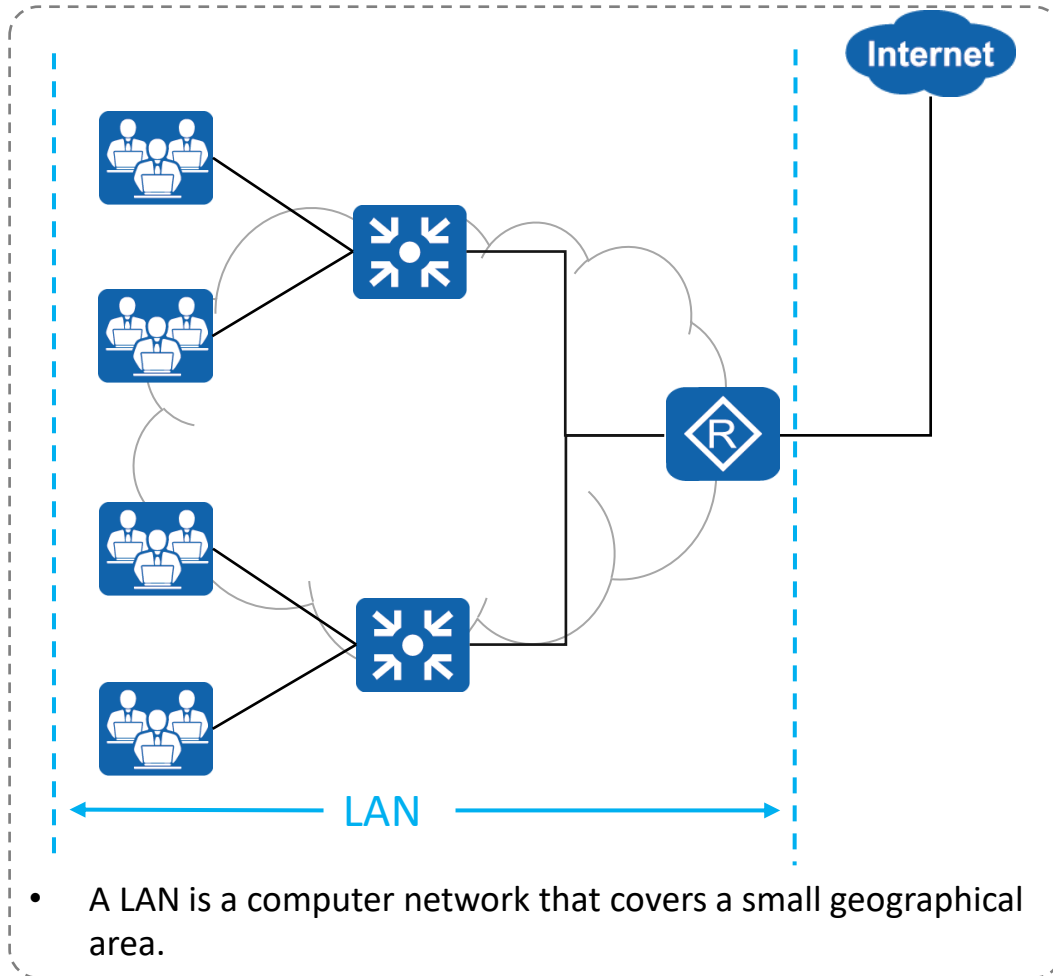
- A **WAN** is a network that connects LANs in different areas. A WAN generally covers tens of kilometers to thousands of kilometers. It can connect multiple regions, cities, and countries, or provide long-distance communication across several continents, forming an international remote network.



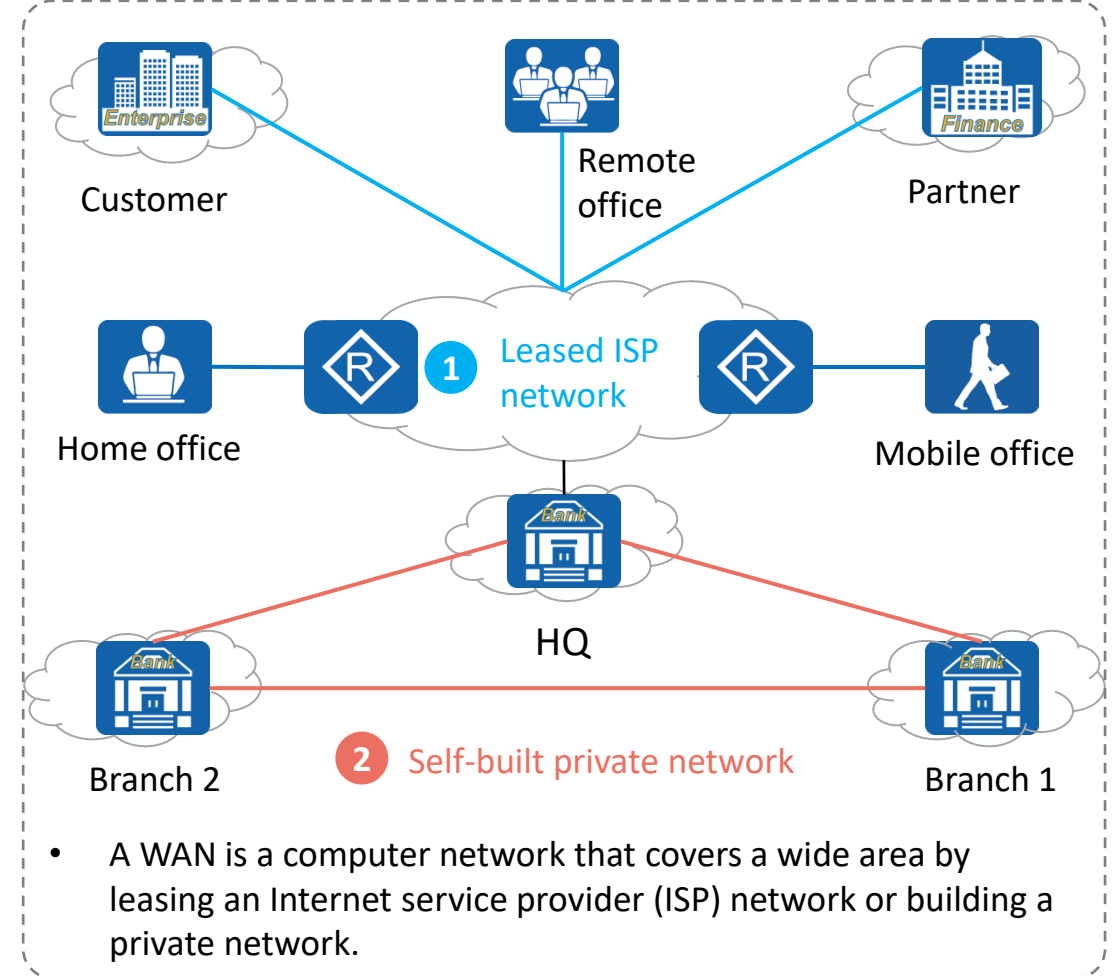


Differences Between a WAN and a LAN

LAN



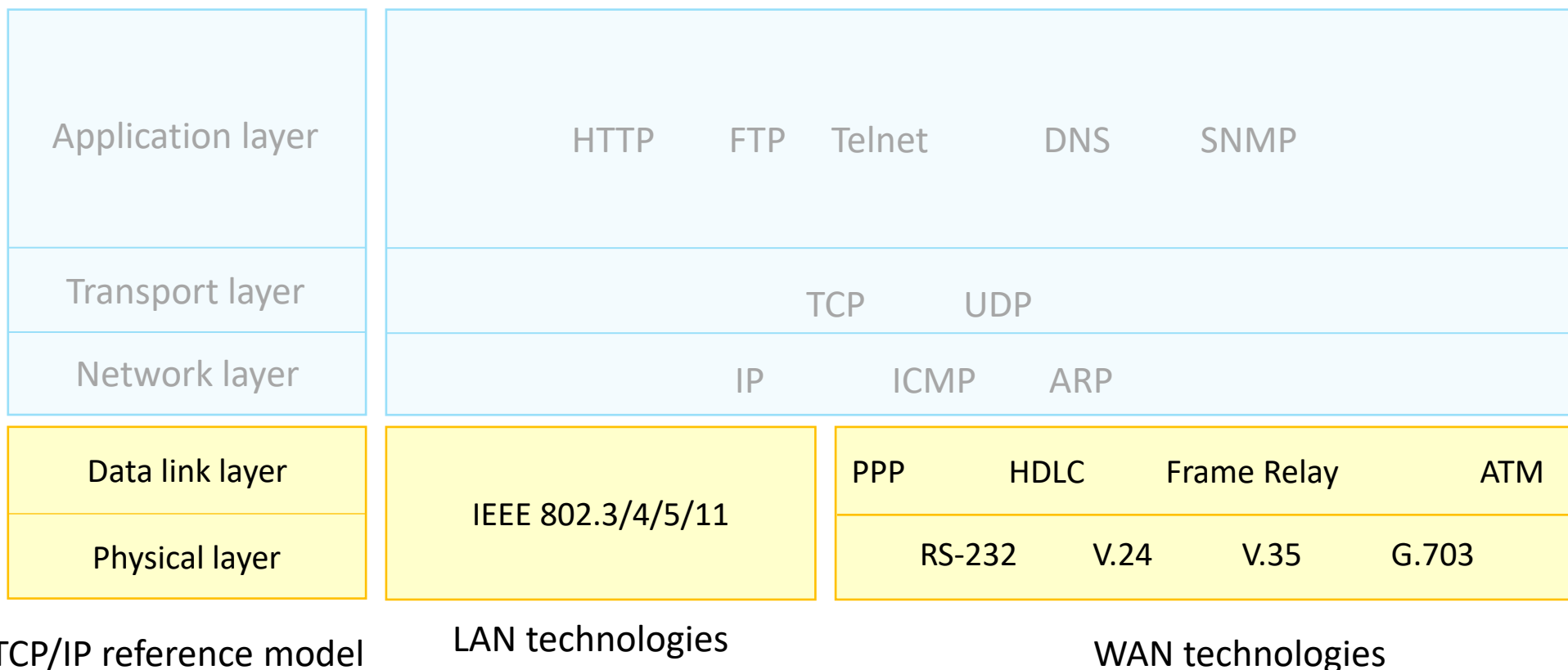
WAN





Overview of Early WAN Technologies

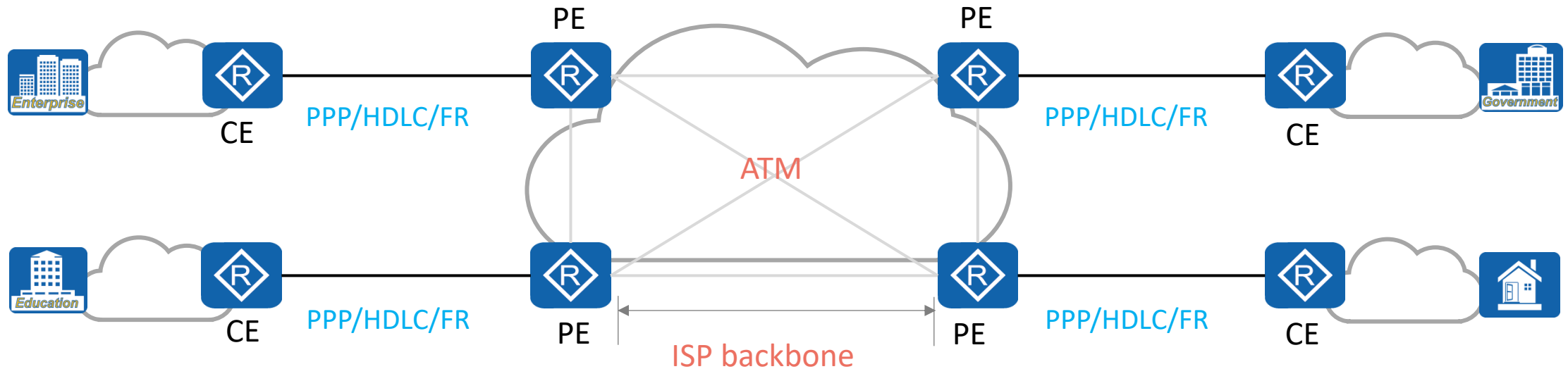
- The early WANs and LANs differ in the data link layer and physical layer and are the same in the other layers in the TCP/IP reference model.





Application of Early WAN Technologies

- The early WAN technologies perform different Layer 2 encapsulation at the data link layer for different types of physical links. PPP, HDLC, and FR are commonly used between CEs and PEs to implement long-distance transmission of user access packets over a WAN. ATM is commonly used on ISP backbone networks for high-speed forwarding.





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PPP Introduction

- PPP is a common WAN data link layer protocol. It is used for P2P data encapsulation and transmission on full-duplex links.
- PPP provides the Password Authentication Protocol (PAP) and Challenge Handshake Authentication Protocol (CHAP).
- PPP features high extensibility. For example, PPP can be extended as Point-to-Point Protocol over Ethernet (PPPoE) when PPP packets need to be transmitted over an Ethernet.
- PPP provides the Link Control Protocol (LCP), which is used to negotiate link layer parameters, such as the maximum receive unit (MRU) and authentication mode.
- PPP provides various Network Control Protocols (NCPs), such as IP Control Protocol (IPCP), for negotiation of network layer parameters and better support for network layer protocols.





PPP Link Setup Process

Feature
Introduction

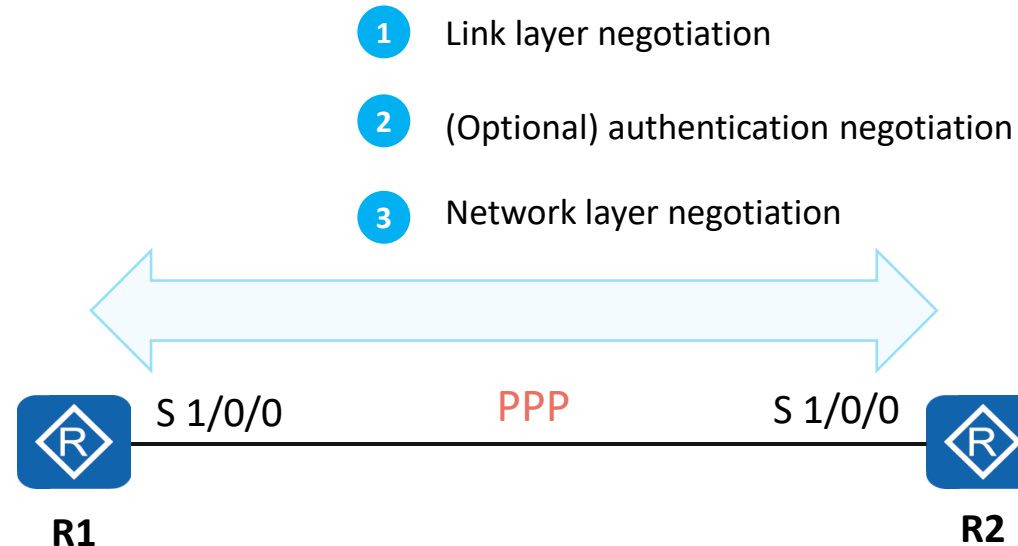
Link Setup

LCP Negotiation

Authentication
Negotiation

NCP Negotiation

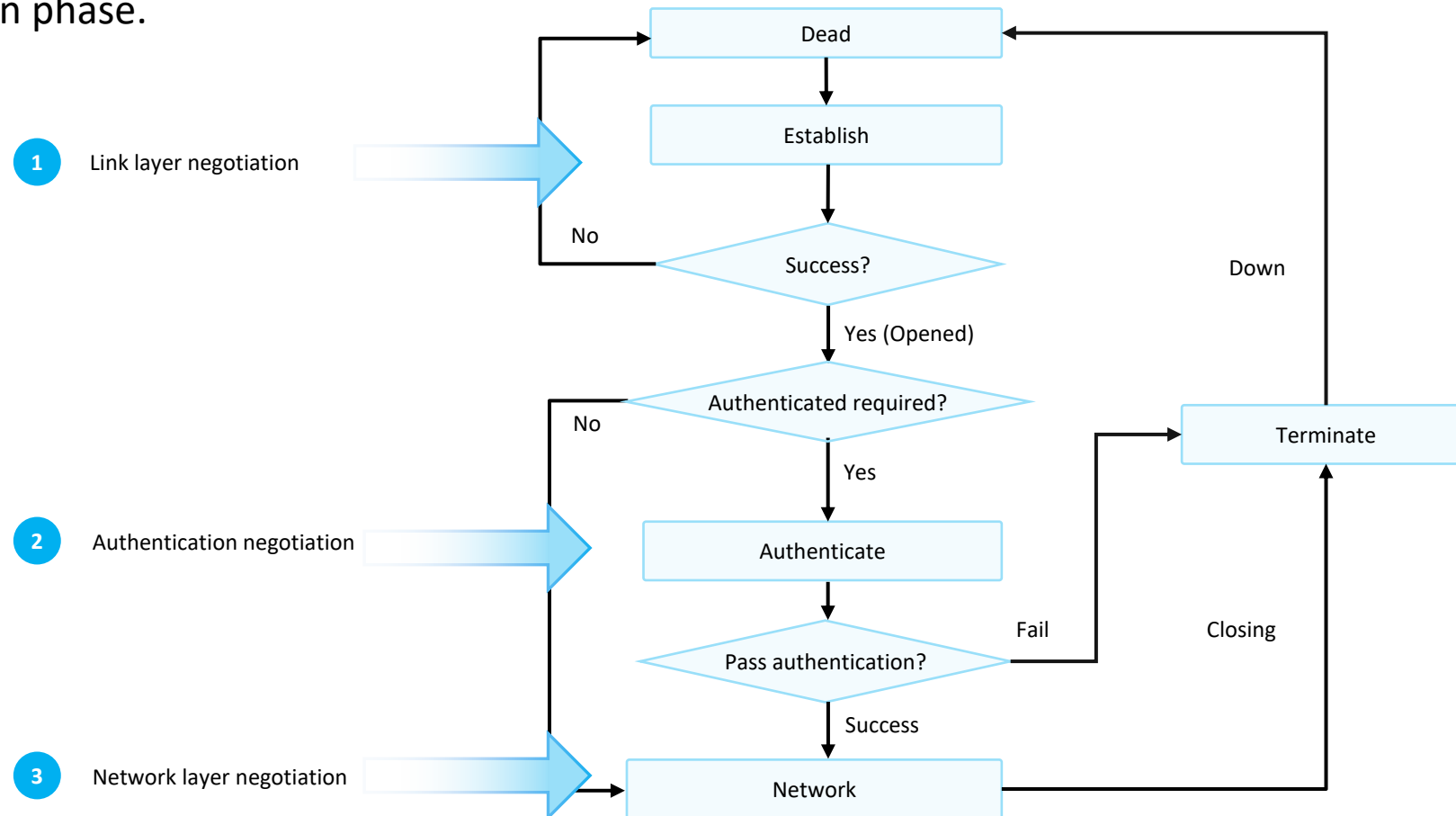
- PPP link setup involves link layer negotiation, optional authentication negotiation, and network layer negotiation.
 - Link layer negotiation: LCP packets are used to negotiate link parameters and establish link layer connections.
 - (Optional) authentication negotiation: The authentication mode negotiated during link layer negotiation is used for link authentication.
 - Network layer negotiation: NCP negotiation is used to select and configure a network layer protocol and negotiate network layer parameters.





State Machine of the PPP Link Interface

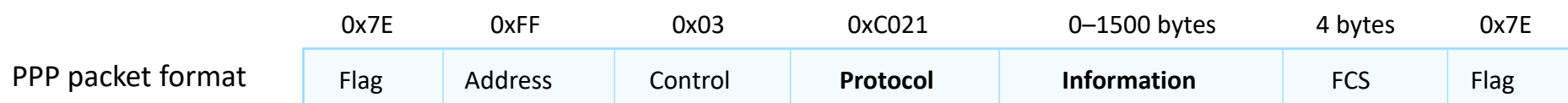
- PPP negotiation is performed by the interfaces at both ends of a link. The interface status indicates the protocol negotiation phase.





LCP Packet Format

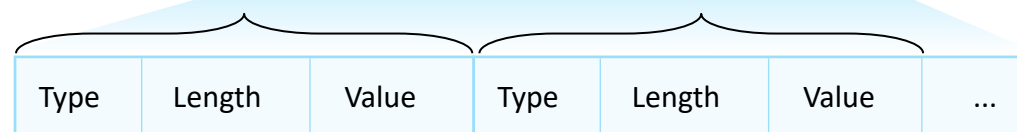
- The Protocol field in a PPP packet identifies the type of the PPP packet. For example, if the Protocol field is 0xC021, the packet is an LCP packet. The Code field is further used to identify different types of LCP packets, as shown in the following table.



0x0021: IP packet
0x8021: IPCP packet
0xC021: LCP packet
0xC023: PAP packet
0xC223: CHAP packet



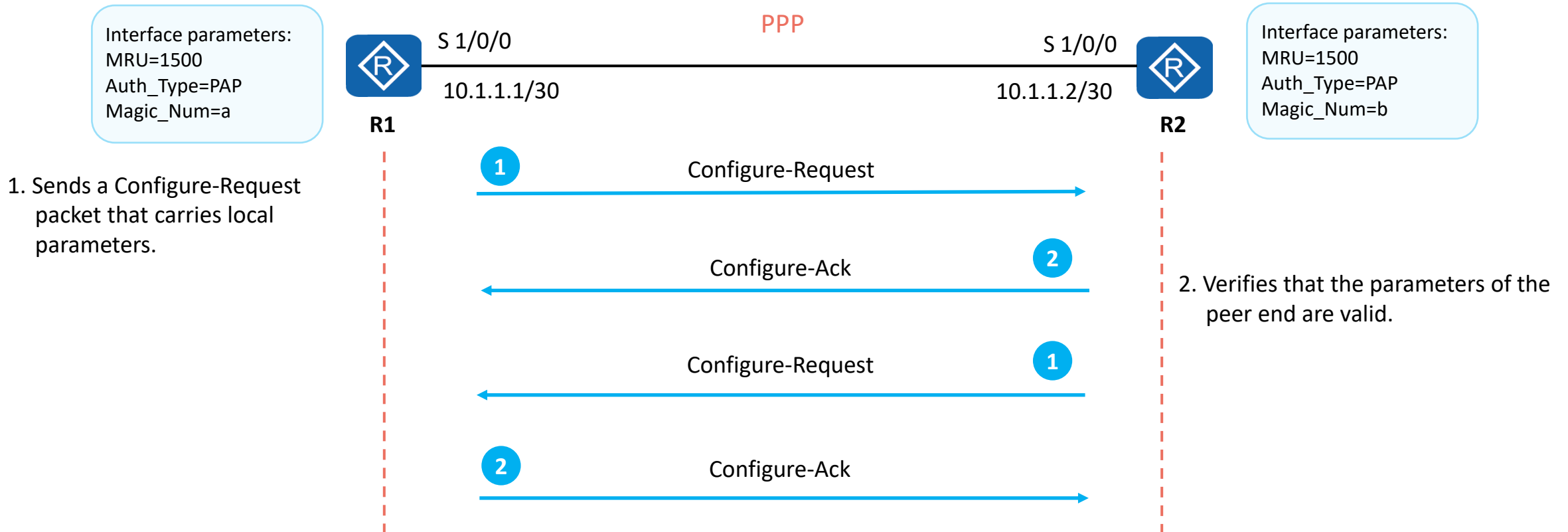
Code	Name	Content
0x01	Configure-Request	Configuration request packet.
0x02	Configure-Ack	Configuration success packet.
0x03	Configure-Nak	Configuration parameters need to be negotiated.
0x04	Configure-Reject	Configuration parameters cannot be identified.



The TLV structure contains common parameters used in LCP negotiation, such as the MRU, authentication protocol, and magic number.

LCP Negotiation Process - Normal Negotiation

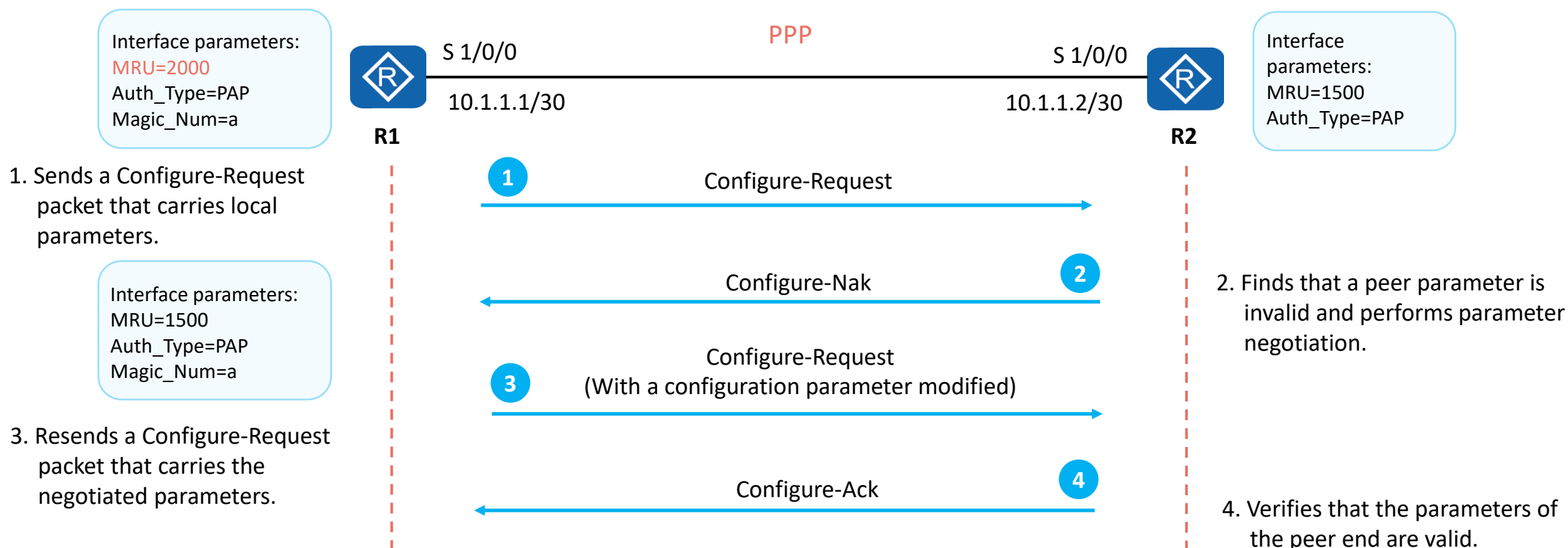
- LCP negotiation is implemented by exchanging different LCP packets. The negotiation is initiated by sending a Configure-Request packet from either party. If the peer end identifies and accepts all parameters in the packet, the peer end returns a Configure-Ack packet to the local end, indicating that the negotiation is successful.





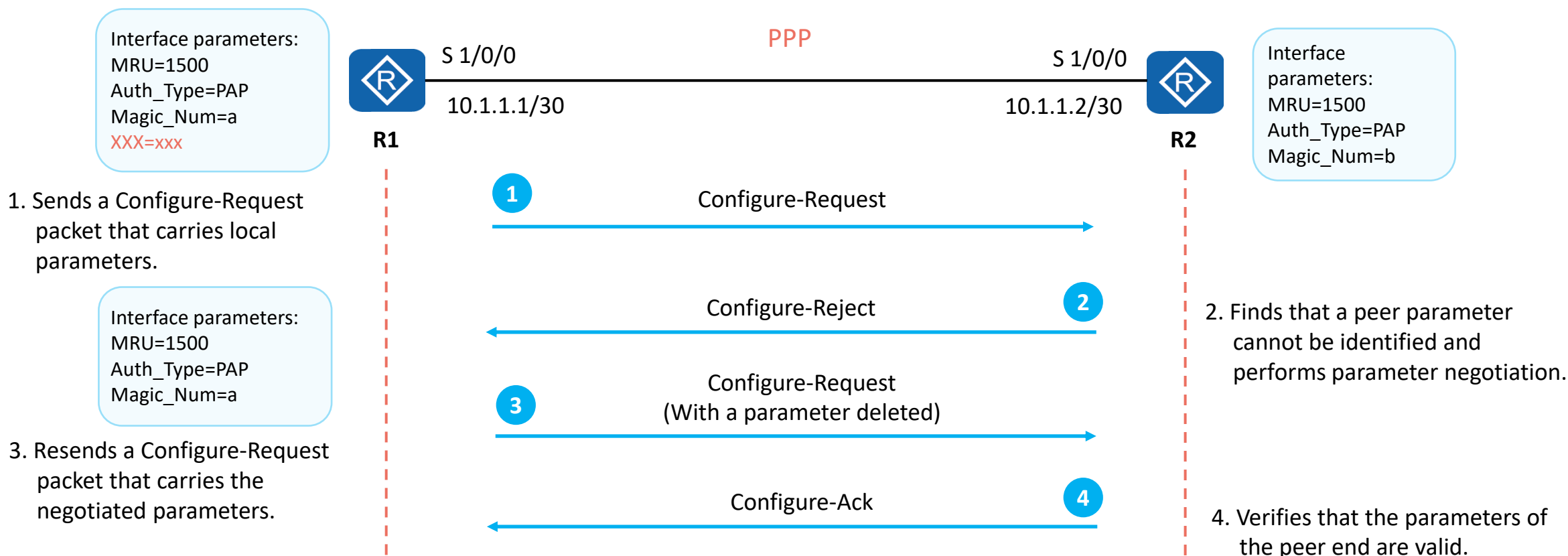
LCP Negotiation Process - Parameter Mismatch

- If LCP parameters do not match during LCP packet exchange, the receiver responds with a Configure-Nak packet to instruct the peer end to modify parameters and perform renegotiation.



LCP Negotiation - Unrecognized Parameters

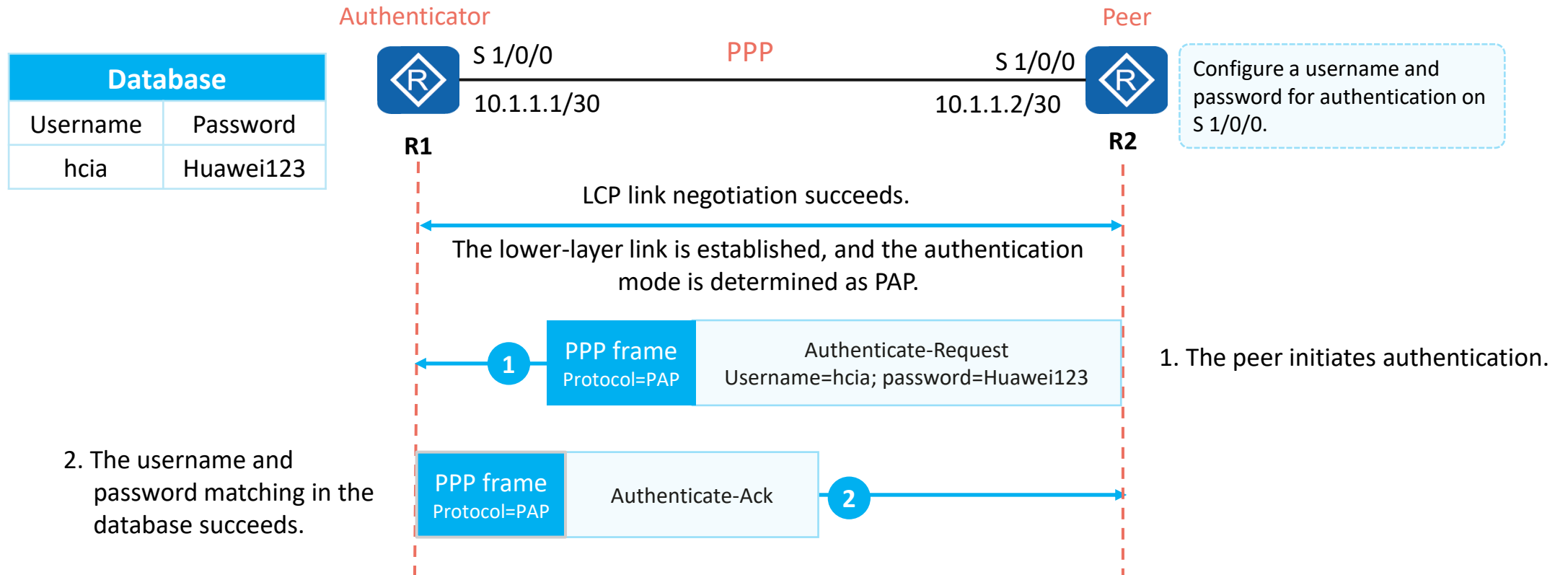
- If LCP parameters cannot be identified during LCP packet exchange, the receiver responds with a Configure-Reject packet to instruct the peer end to delete the unidentifiable parameters and renegotiates with the peer end.





PPP Authentication Mode - PAP

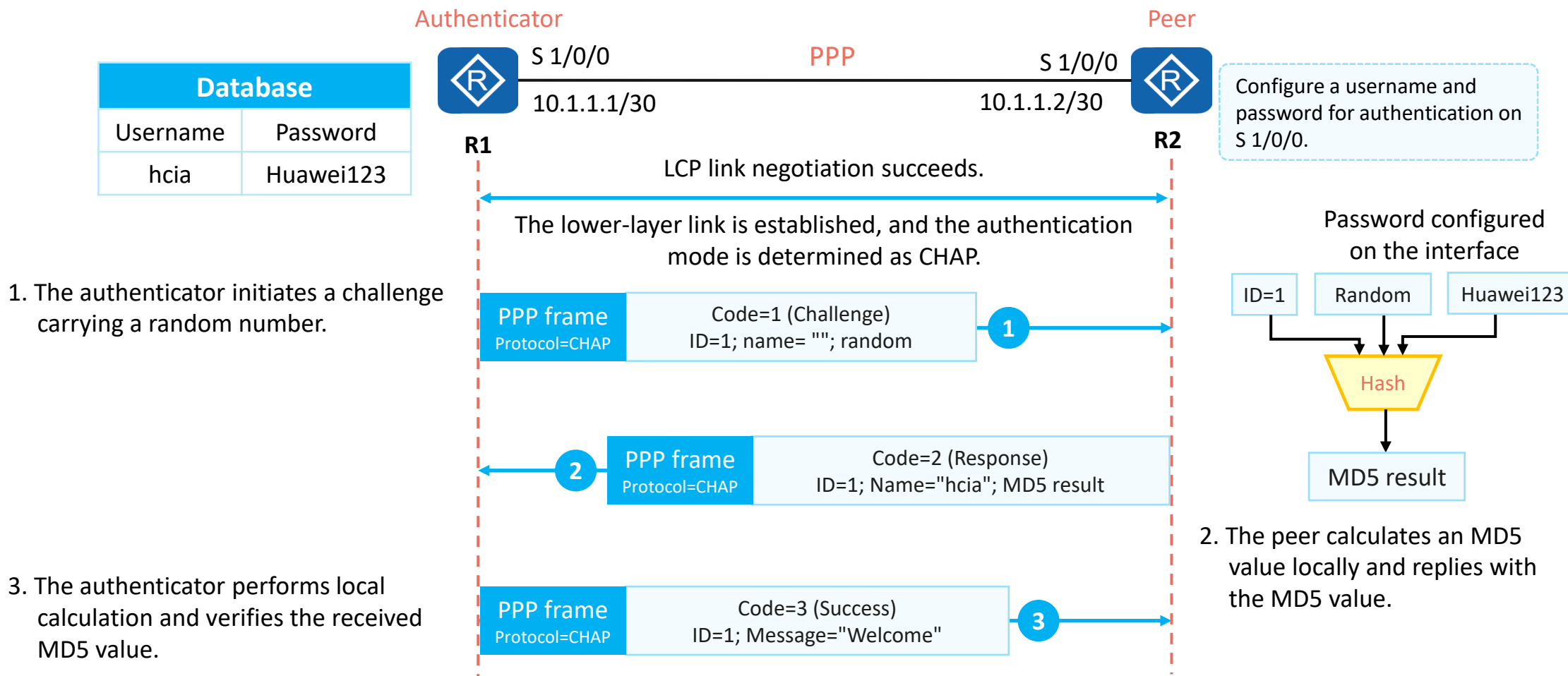
- After the link negotiation is successful, authentication negotiation can be performed. There are two authentication negotiation modes: PAP and CHAP.
- PAP authentication requires a two-way handshake. Negotiation packets are transmitted on the link in clear text.





PPP Authentication Mode - CHAP

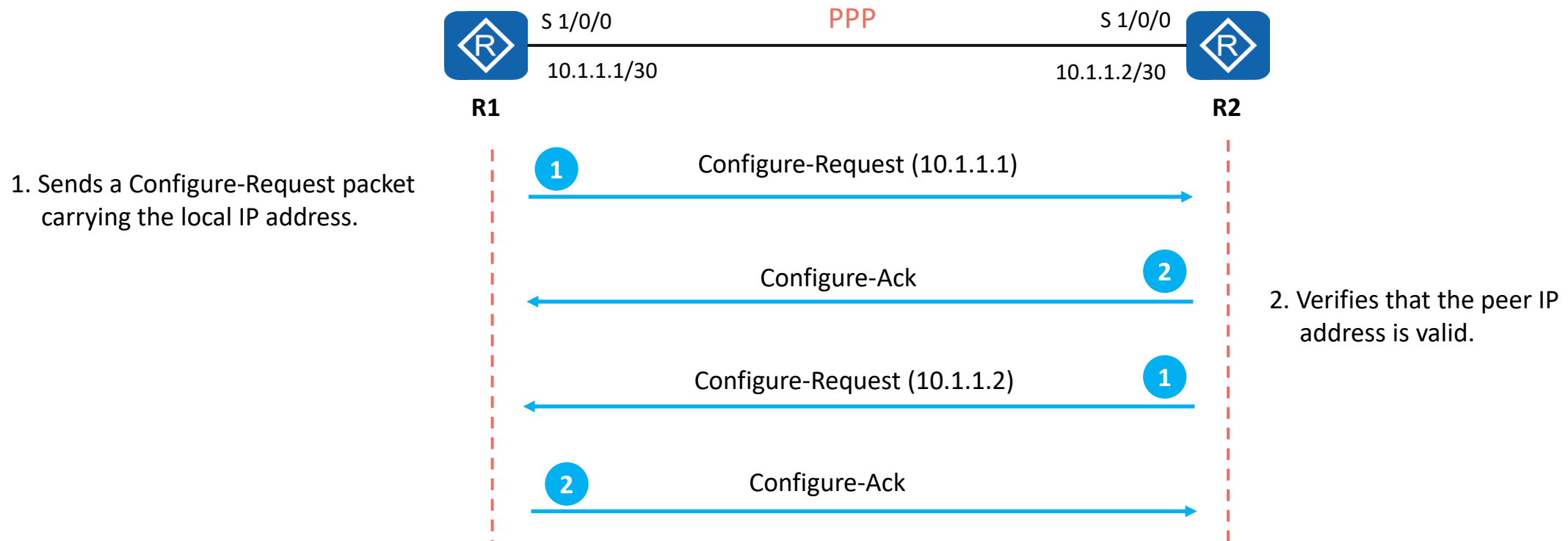
- CHAP authentication requires a three-way handshake. Negotiation packets are encrypted before being transmitted on a link.





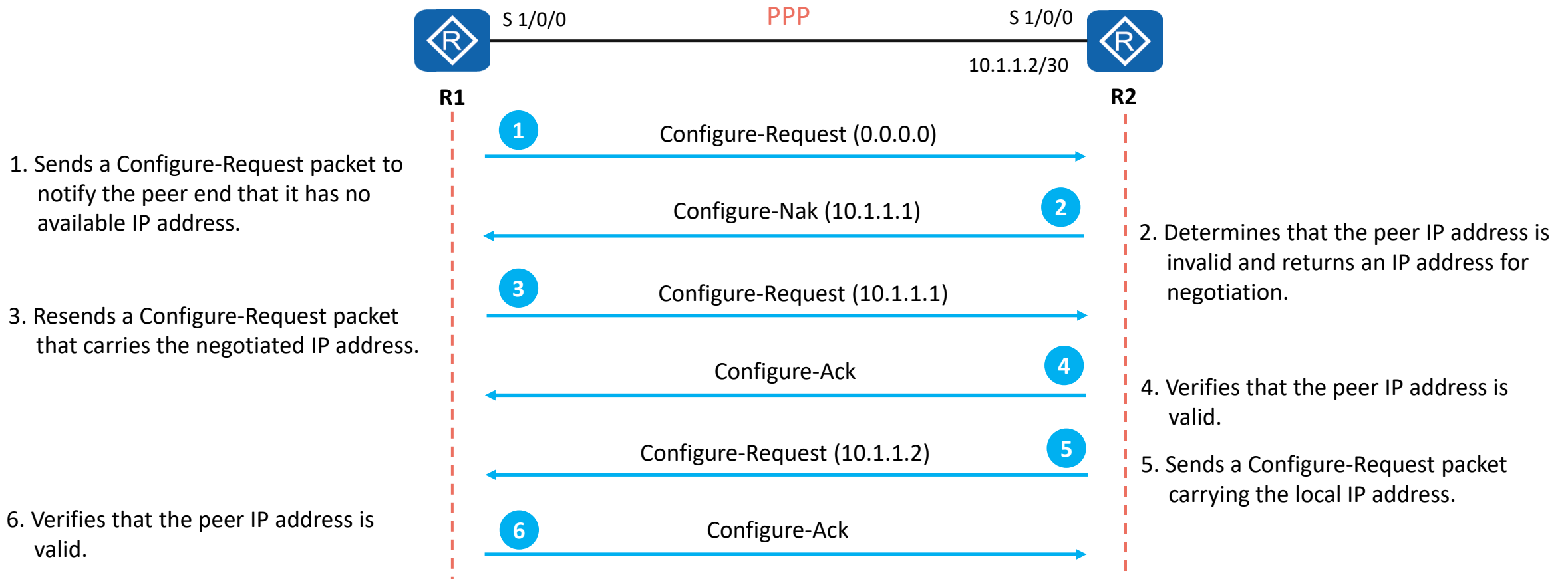
NCP Negotiation - Static IP Address Negotiation

- After PPP authentication negotiation, the two ends enter the NCP negotiation phase to negotiate the format and type of data packets transmitted on the data link. IPCP, for example, is classified into static and dynamic IP address negotiation.
- Static IP address negotiation requires manual configuration of IP addresses at both ends of a link.



NCP Negotiation - Dynamic IP Address Negotiation

- In dynamic IP address negotiation, one end of a PPP link can assign an IP address to the other end.





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Configuring Basic PPP Functions

1. Encapsulate an interface with PPP.

```
[Huawei-Serial0/0/0] link-protocol ppp
```

In the interface view, change the interface encapsulation protocol to PPP. The default encapsulation protocol of Huawei devices' serial interfaces is PPP.

2. Configure a negotiation timeout period.

```
[Huawei-Serial0/0/0] ppp timer negotiate seconds
```

During LCP negotiation, the local end sends an LCP negotiation packet to the peer end. If the local end does not receive a reply packet from the peer end within the specified negotiation timeout period, the local end resends an LCP negotiation packet.



Configuring PAP Authentication

1. Configure an authenticator to authenticate a peer using the PAP mode.

```
[Huawei-aaa] local-user user-name password { cipher | irreversible-cipher } password  
[Huawei-aaa] local-user user-name service-type ppp
```

```
[Huawei-Serial0/0/0] ppp authentication-mode pap
```

Before configuring the authenticator to authenticate a peer using the PAP mode, add the username and password of the peer to the local user list in the AAA view. Then select the PAP authentication mode.

2. Configure the peer to be authenticated by the authenticator in PAP mode.

```
[Huawei-Serial0/0/0] ppp pap local-user user-name password { cipher | simple } password
```

This command configures the peer to send its username and password to the authenticator.



Configuring CHAP Authentication

1. Configure an authenticator to authenticate a peer using CHAP mode.

```
[Huawei-aaa] local-user user-name password { cipher | irreversible-cipher } password  
[Huawei-aaa] local-user user-name service-type ppp
```

```
[Huawei-Serial0/0/0] ppp authentication-mode chap
```

2. Configure the peer to be authenticated by the authenticator in CHAP mode.

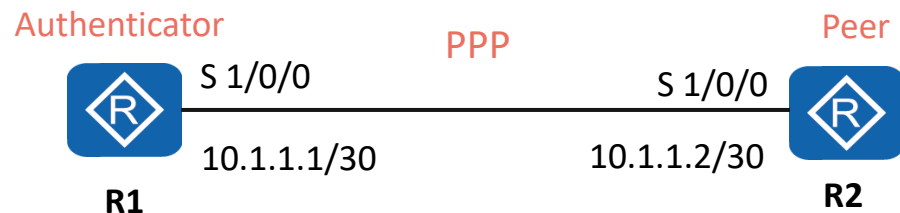
```
[Huawei-Serial0/0/0] ppp chap user user-name
```

```
[Huawei-Serial0/0/0] ppp chap password { cipher | simple } password
```

This command configures a local username and a password for CHAP authentication.



Example for Configuring PAP Authentication



- Experiment requirements:
 1. Enable PAP authentication on the PPP link between R1 and R2.
 2. Configure R1 as the authenticator.
 3. Configure R2 as the peer.

Configurations on R1

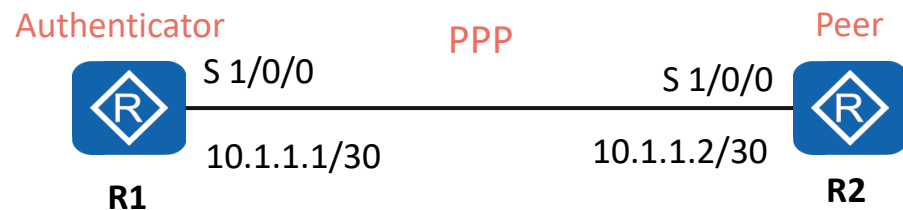
```
[R1]aaa # Add information about the user to be authenticated.  
[R1-aaa]local-user huawei password cipher huawei123  
[R1-aaa]local-user huawei service-type ppp  
# Specify the service type of the user to be authenticated.  
[R1]interface Serial 1/0/0  
[R1-Serial1/0/0]link-protocol ppp  
[R1-Serial1/0/0]ppp authentication-mode pap  
# Set the authentication mode to PAP.  
[R1-Serial1/0/0]ip address 10.1.1.1 30
```

Configurations on R2

```
[R2]interface Serial 1/0/0  
[R2-Serial1/0/0]link-protocol ppp  
[R2-Serial1/0/0]ppp pap local-user huawei password cipher huawei123 #  
Add user information for PPP authentication.  
[R2-Serial1/0/0]ip address 10.1.1.2 30
```



Example for Configuring CHAP Authentication



- Experiment requirements:
 1. Enable CHAP authentication on the PPP link between R1 and R2.
 2. Configure R1 as the authenticator.
 3. Configure R2 as the peer.

Configurations on R1

```
[R1]aaa # Add information about the user to be authenticated.  
[R1-aaa]local-user huawei password cipher huawei123  
[R1-aaa]local-user huawei service-type ppp  
# Specify the service type of the user to be authenticated.  
[R1]interface Serial 1/0/0  
[R1-Serial1/0/0]link-protocol ppp  
[R1-Serial1/0/0]ppp authentication-mode chap  
# Set the authentication mode to CHAP.
```

Configurations on R2

```
[R2]interface Serial 1/0/0  
[R2-Serial1/0/0]link-protocol ppp  
[R2-Serial1/0/0]ppp chap user huawei  
[R2-Serial1/0/0]ppp chap password cipher huawei123  
# Add user information for PPP authentication.
```



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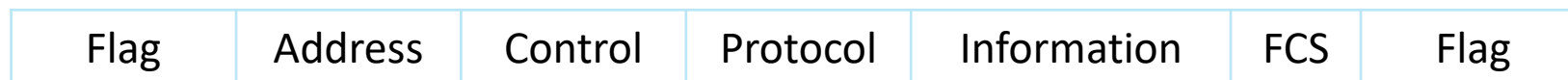
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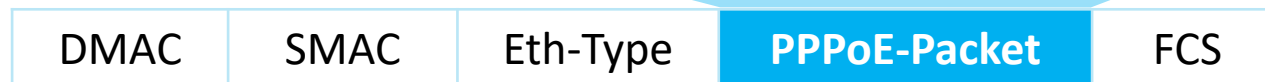
What Is PPPoE?

- PPP over Ethernet (PPPoE) is a link layer protocol that encapsulates PPP frames into Ethernet frames. PPPoE enables multiple hosts on an Ethernet to connect to a broadband remote access server (BRAS).
- PPPoE integrates the advantages of Ethernet and PPP. It has the flexible networking advantage of Ethernet and can use PPP to implement authentication and accounting.

PPP frame structure



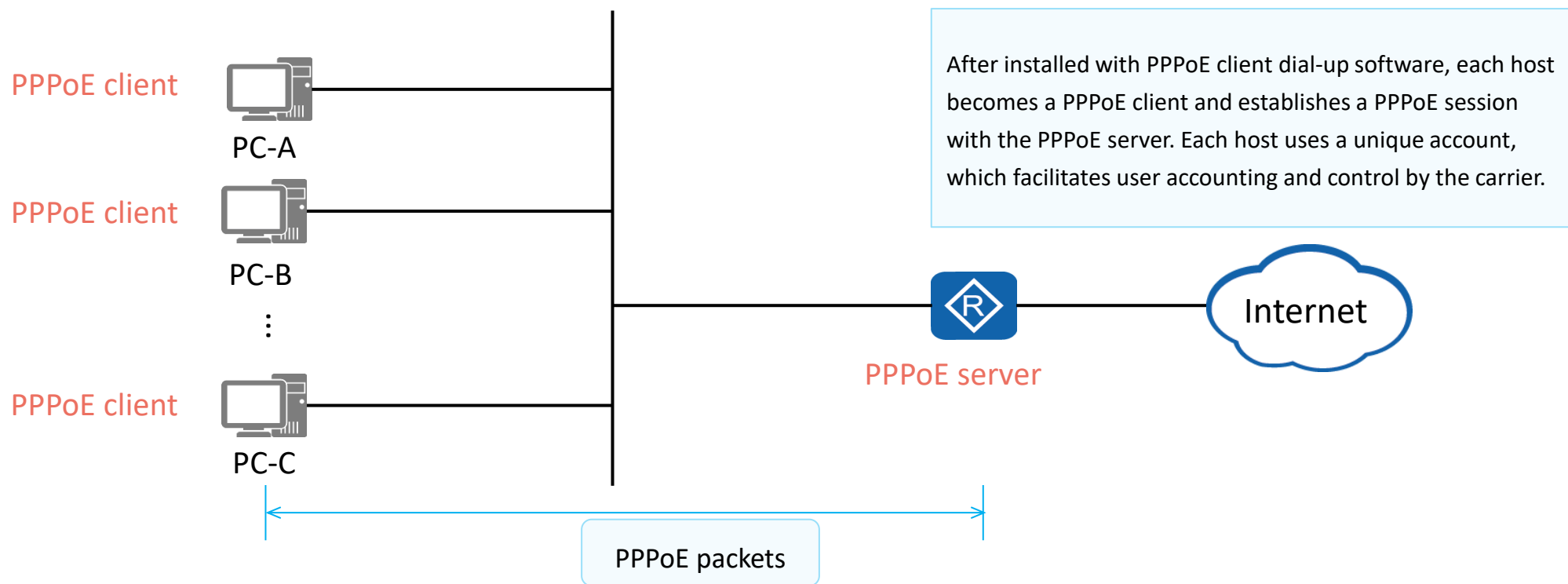
PPPoE frame structure





PPPoE Application Scenarios

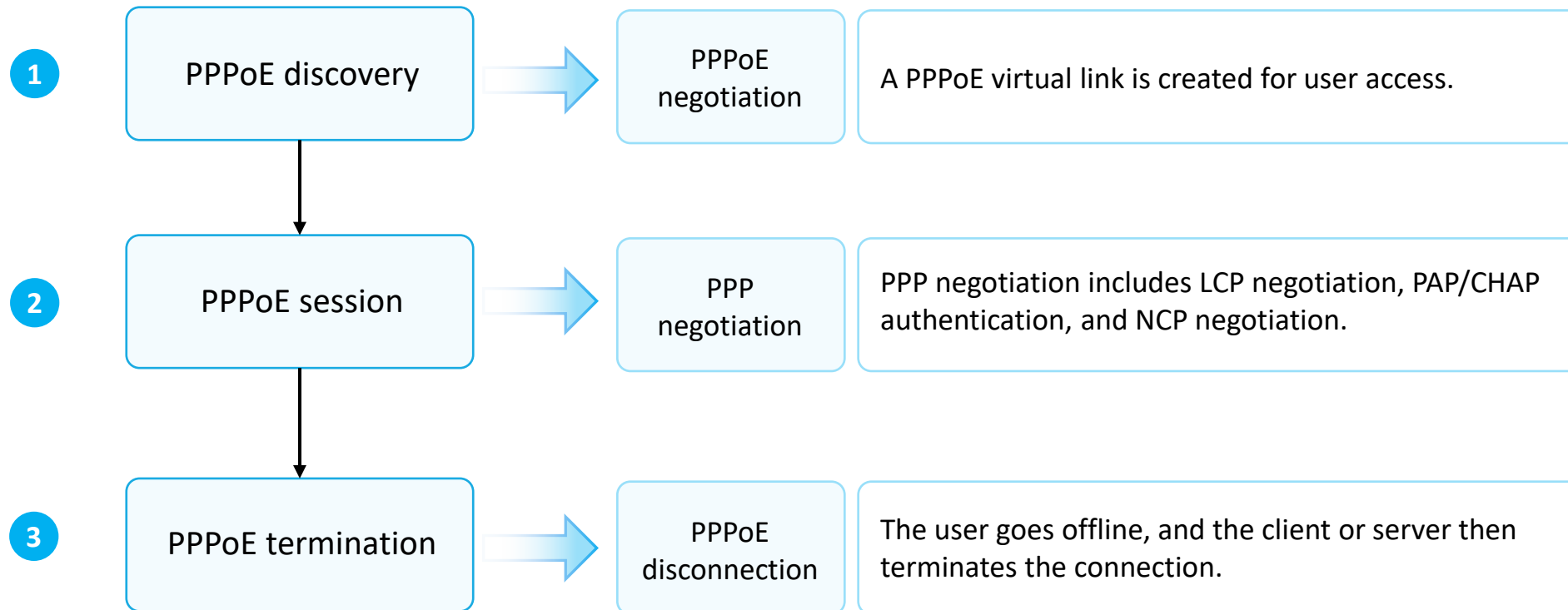
- PPPoE provides P2P connections on an Ethernet. A PPPoE client and a PPPoE server establish a PPP session to encapsulate PPP data packets and provide access services for hosts on the Ethernet, implementing user control and accounting. PPPoE is widely used on enterprise and carrier networks.
- PPPoE is usually used by home users and enterprise users to dial up to access the Internet.





PPPoE Session Establishment

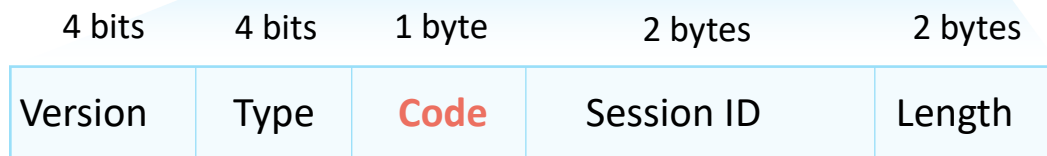
- PPPoE session establishment involves three stages: PPPoE discovery, session, and termination stages.





PPPoE Packets

- A PPPoE session is established by exchanging different PPPoE packets. The PPPoE packet structure and common packet types are as follows.



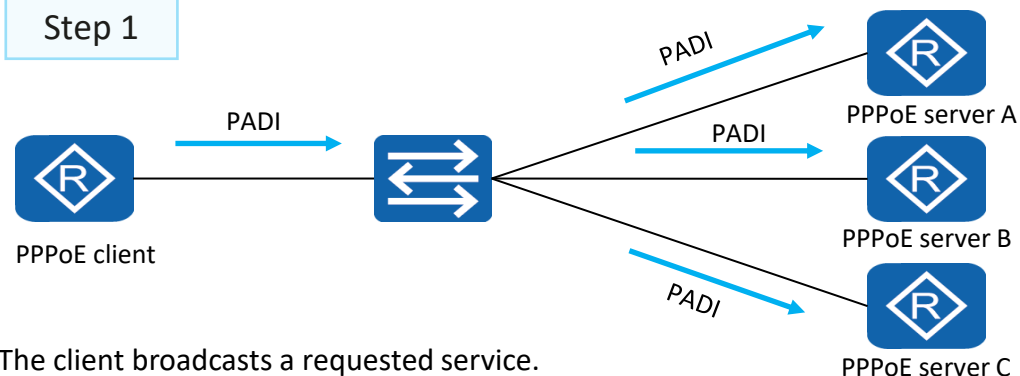
Code	Name	Content
0x09	PADI	PPPoE Active Discovery Initiation packet
0x07	PADO	PPPoE Active Discovery Offer packet
0x19	PADR	PPPoE Active Discovery Request packet
0x65	PADS	PPPoE Active Discovery Session-confirmation packet
0xa7	PADT	PPPoE Active Discovery Terminate packet



PPPoE Discovery Stage

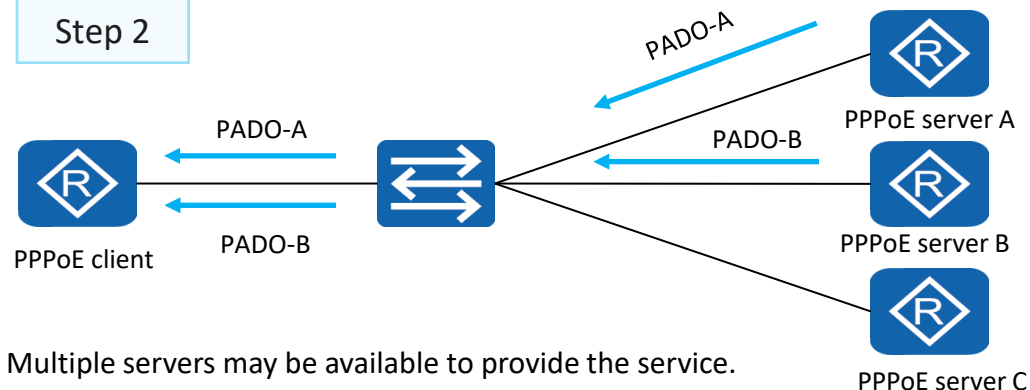
- PPPoE discovery involves four steps: 1) the client sends a request, 2) the servers respond to the request, 3) the client confirms a response and 4) establishes a session.

Step 1



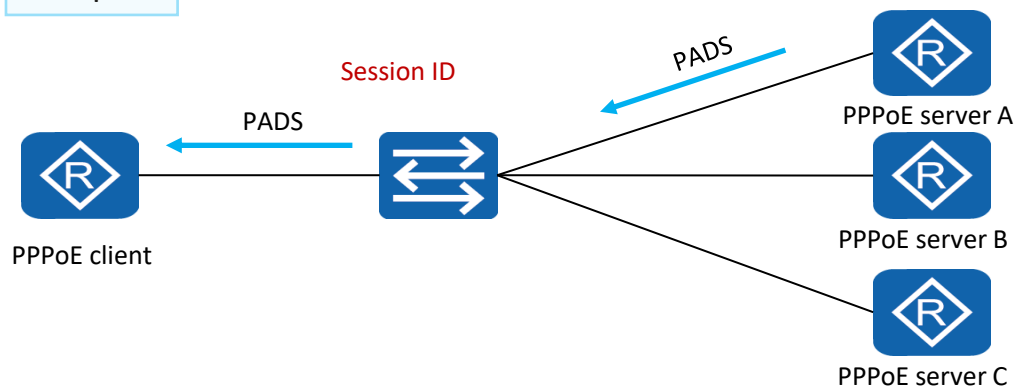
- The client broadcasts a requested service.

Step 2



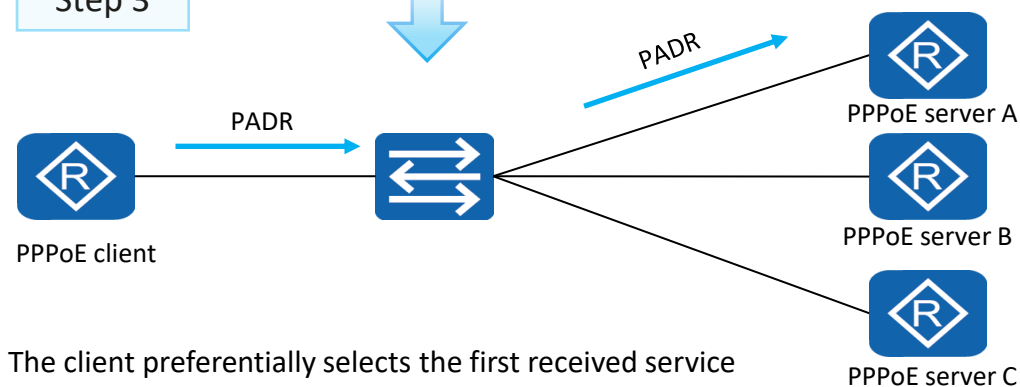
- Multiple servers may be available to provide the service.

Step 4



- The server assigns a session ID to the client to establish a session.

Step 3

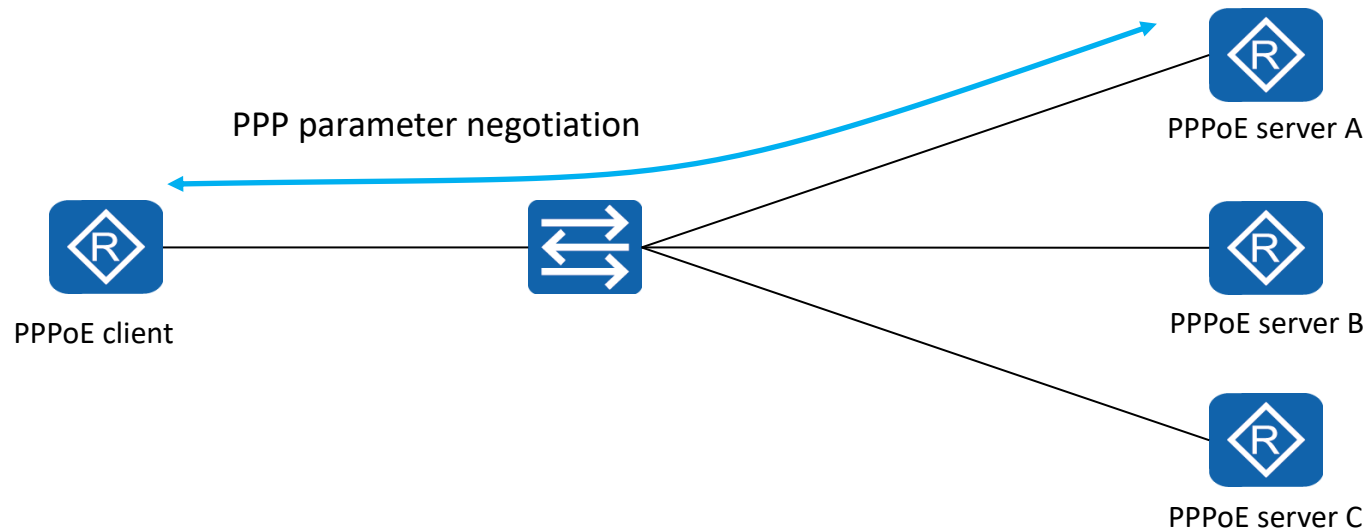


- The client preferentially selects the first received service response and sends a service request.



PPPoE Session Stage

- In the PPPoE session stage, PPP negotiation, including LCP, authentication, and NCP negotiation, is performed.

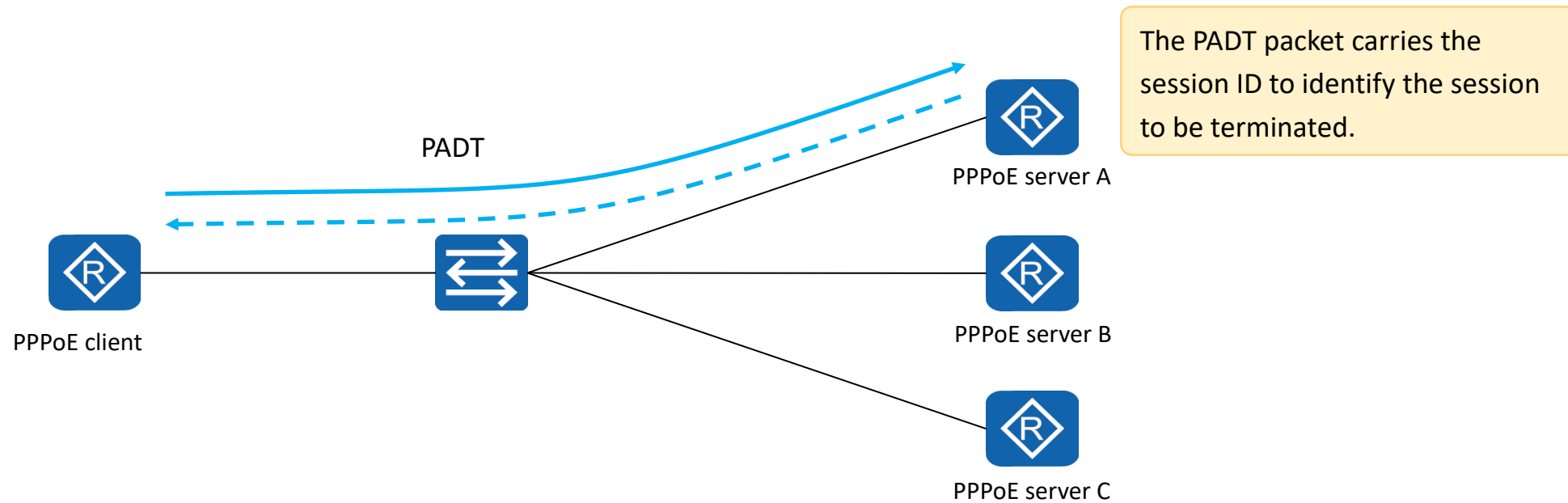


In the entire session stage, the session ID allocated by the PPPoE server remains unchanged.



PPPoE Session Termination Stage

- If the PPPoE client wants to terminate the session, it sends a PADT packet to the PPPoE server.
- Similarly, if the PPPoE server wants to terminate the session, it sends a PADT packet to the PPPoE client.





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Configuring Basic PPPoE Functions

1. Configure a dialer rule and set conditions for initiating a PPPoE session under the rule.

```
[Huawei] dialer-rule
```

Configure a username on the dialer interface. The username must be the same as that of the peer server.

```
[Huawei-Dialer1] dialer user username
```

3. Add the interface to a dialer group.

```
[Huawei-Dialer1] dialer-group group-number
```

4. Specify a dialer bundle for the interface.

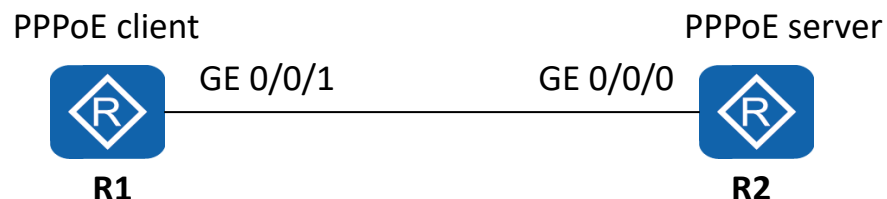
```
[Huawei-Dialer1] dialer-bundle number
```

5. Bind a physical interface to the dialer bundle.

```
[Huawei-Ethernet0/0/0] pppoe-client dial-bundle-number number
```



Example for Configuring a PPPoE Client (1)



- Experiment requirements:

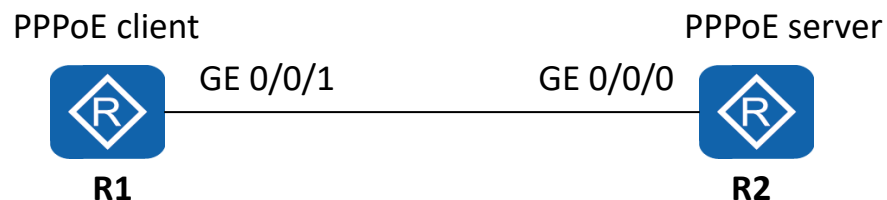
1. Configure R1 as a PPPoE client and R2 as a PPPoE server.
2. Configure a dialer interface for the PPPoE client on R1.
3. Configure the authentication function on the dialer interface on R1.
4. The dialer interface on R1 can obtain the IP address allocated by the PPPoE server.
5. R1 can access the server through the dialer interface.

1. Create a dialer interface and configure a username and password for authentication.

```
[R1]dialer-rule
[R1-dialer-rule]dialer-rule 1 ip permit
[R1-dialer-rule]quit
[R1]interface dialer 1
[R1-Dialer1] dialer user enterprise
[R1-Dialer1] dialer-group 1
[R1-Dialer1] dialer bundle 1
[R1-Dialer1] ppp chap user huawei1
[R1-Dialer1] ppp chap password cipher huawei123
[R1-Dialer1] ip address ppp-negotiate
```



Example for Configuring a PPPoE Client (2)



- Experiment requirements:

1. Configure R1 as a PPPoE client and R2 as a PPPoE server.
2. Configure a dialer interface for the PPPoE client on R1.
3. Configure the authentication function on the dialer interface on R1.
4. The dialer interface on R1 can obtain the IP address allocated by the PPPoE server.
5. R1 can access the server through the dialer interface.

2. Bind the dialer interface to an outbound interface.

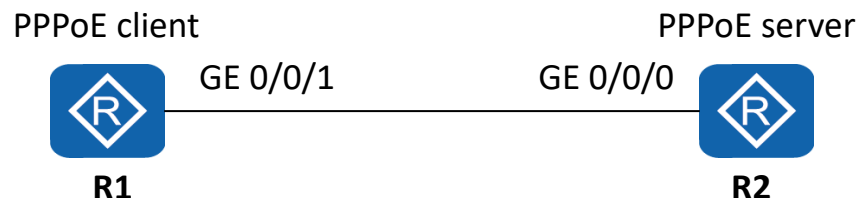
```
[R1]interface GigabitEthernet 0/0/1
[R1-GigabitEthernet0/0/1]pppoe-client dial-bundle-number 1
[R1-GigabitEthernet0/0/1]quit
```

3. Configure a default route from the PPPoE client to the server.

```
[R1]ip route-static 0.0.0.0 0.0.0.0 dialer 1
```



Example for Configuring a PPPoE Server



- Experiment requirements:
 - Create an address pool on the PPPoE server for address allocation to the PPPoE client.
 - The PPPoE server authenticates the PPPoE client and assigns a valid IP address to the client.

1. Create an address pool and a virtual template.

```
[R2]ip pool pool1 # Create an address pool and specify the range of the IP
addresses to be allocated and a gateway.
[R2-ip-pool-pool1]network 192.168.1.0 mask 255.255.255.0
[R2-ip-pool-pool1]gateway-list 192.168.1.254
[R2]interface Virtual-Template 1 # Create a virtual template interface.
[R2-Virtual-Template1]ppp authentication-mode chap
[R2-Virtual-Template1]ip address 192.168.1.254 255.255.255.0
[R2-Virtual-Template1]remote address pool pool1
```

2. Bind a physical interface to the virtual template.

```
[R2]interface GigabitEthernet 0/0/0
[R2-GigabitEthernet0/0/0]pppoe-server bind virtual-template 1
[R2-GigabitEthernet0/0/0]quit
```

3. Create an access user.

```
[R2]aaa # Add information about the user to be authenticated.
[R2-aaa]local-user huawei1 password cipher huawei123
[R2-aaa]local-user huawei1 service-type ppp
```




Verifying the Configuration

1. Check detailed information about the dialer interface.

```
<R1>display interface Dialer 1
Dialer1 current state: UP
Line protocol current state: UP (spoofing)
Description: HUAWEI, AR Series, Dialer1 Interface
Route Port, The Maximum Transmit Unit is 1500, Hold timer is
10(sec)
Internet Address is negotiated, 192.168.10.254/32
Link layer protocol is PPP
LCP initial
Physical is Dialer
Bound to Dialer1:0:
Dialer1:0 current state : UP
Line protocol current state : UP
Link layer protocol is PPP
LCP opened, IPCP opened
```

2. Check the initial status of the PPPoE session on the client.

```
[R1]display pppoe-client session summary
PPPoE Client Session:
ID  Bundle  Dialer Intf    Client-MAC    Server-MAC    State
0   1        1  GE0/0/1  54899876830c  000000000000  IDLE
```

3. Check the establishment status of the PPPoE session on the client.

```
[R1]display pppoe-client session summary
PPPoE Client Session:
ID  Bundle  Dialer Intf    Client-MAC    Server-MAC    State
1   1        1  GE0/0/1  00e0fc0308f6  00e0fc036781  UP
```



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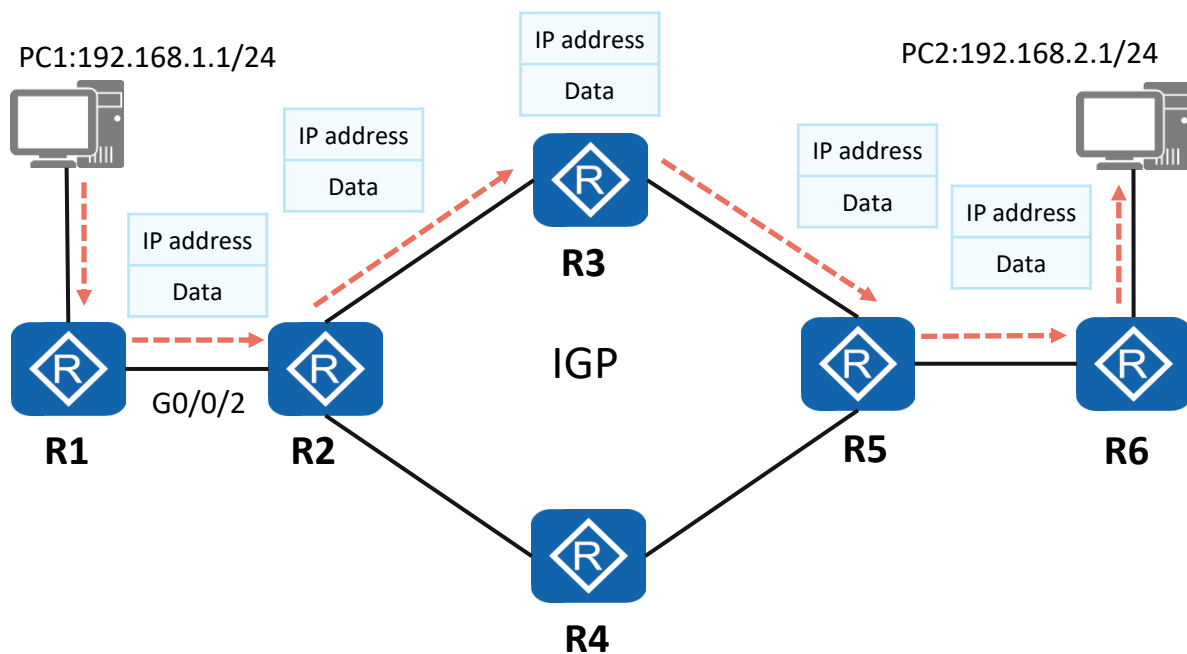
Evolution of WAN Technologies

- The data link layer protocols commonly used on early WANs include PPP, HDLC, and ATM. With the network evolution towards all-IP, the IP-based Internet becomes popular. However, the IP technology based on the longest match rule must use software to search for routes, resulting in low forwarding performance, which has become the bottleneck that restricts the network development.
- Multiprotocol Label Switching (MPLS) was originally proposed to improve the forwarding speeds of routers. Compared with the traditional IP routing mode, MPLS parses IP packet headers only at the network edges during data forwarding. Transit nodes forward packets based on labels, without the need to parse IP packet headers. This speeds up software processing.
- With the improvement of router performance, the route search speed is no longer a bottleneck for network development. Thus, MPLS loses its advantage in fast forwarding speed. However, leveraging support for multi-layer labels and a connection-oriented forwarding plane, MPLS is widely applied in various scenarios, such as virtual private network (VPN), traffic engineering (TE), and quality of service (QoS) scenarios.



Traditional IP Routing and Forwarding

- Traditional IP forwarding uses hop-by-hop forwarding. Each time a data packet passes through a router, the router decapsulates the packet to check the network layer information and searches its routing table based on the longest match rule to guide packet forwarding. The repeat process of decapsulating packets, searching routing tables, and re-encapsulating the packets on routers lead to low forwarding performance.



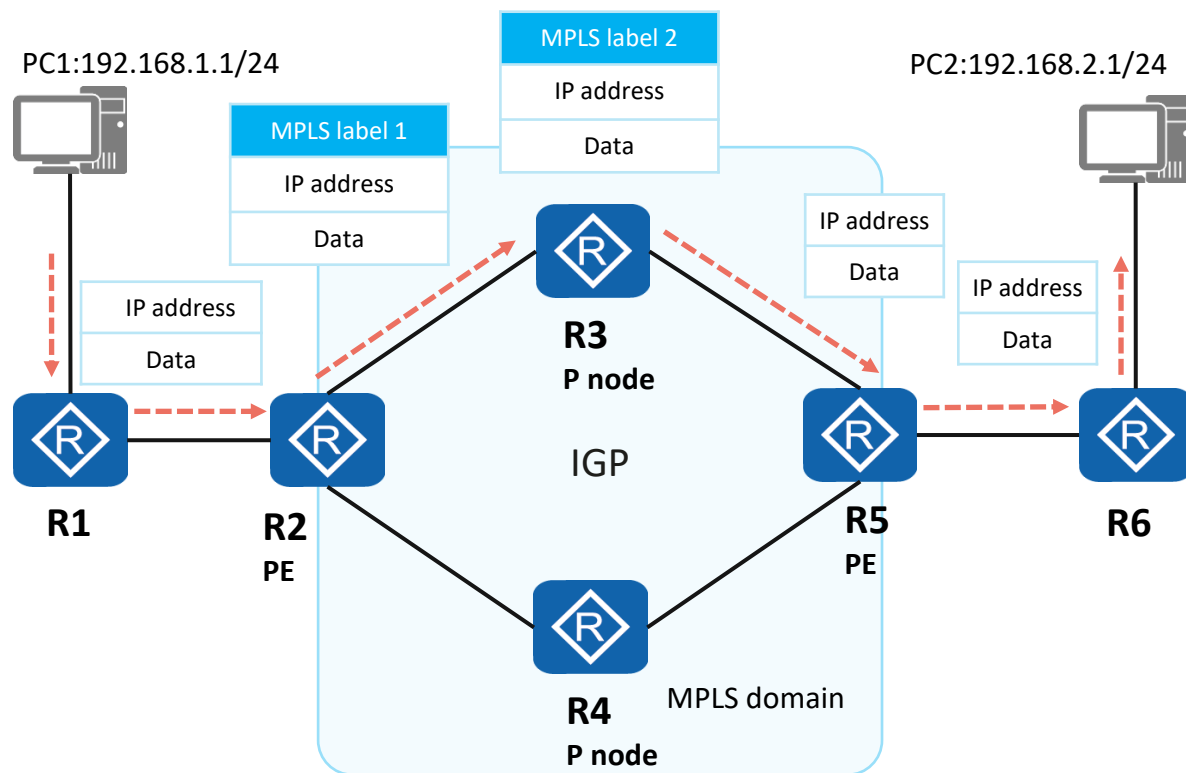
- Characteristics of traditional IP routing and forwarding:
 - All routers need to know the network-wide routes.
 - Traditional IP forwarding is connectionless-oriented and cannot provide good end-to-end QoS guarantee.

R1 routing table

Destination/Mask	Protocol	Preference	Cost	NextHop	Interface
192.168.1.0/24	Direct	0	0	192.168.1.254	GE 0/0/0
192.168.12.0/24	Direct	0	0	192.168.12.1	GE 0/0/2
192.168.2.0/24	OSPF	10	3	192.168.12.2	GE 0/0/2



MPLS Label-based Forwarding

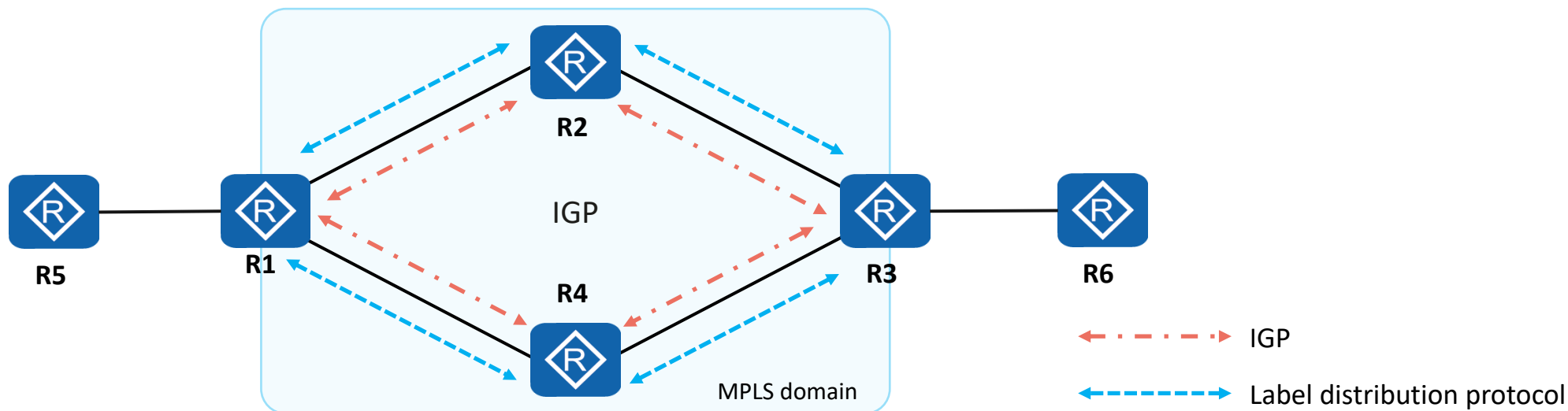


- MPLS is used on IP backbone networks.
- MPLS is a tunneling technology that provides connection-oriented switching for the network layer based on IP routing and control protocols. It provides better QoS guarantee.
- MPLS labels, instead of IP routes, are searched for to forward packets, which greatly improves forwarding efficiency.
- Labels used in MPLS forwarding can be manually configured or dynamically allocated using a label distribution protocol.



MPLS Forwarding Problems

- MPLS labels can be statically or dynamically distributed. The involved problems are as follows:
 - Static label distribution requires manual configuration. As the network scale expands, network topologies are prone to change. Static label configuration cannot meet the requirements of large-scale networks.
 - Some dynamic label distribution protocols do not have the path computation capability and need to use IGPs to compute paths. In addition, the control planes of these protocols are complex, requiring devices to send a large number of messages to maintain peer and path status, wasting link bandwidth and device resources. What is more, despite supporting TE, some label distribution protocols require complex configurations and do not support load balancing. Devices have to send a large number of protocol packets to maintain proper paths. In addition, as devices are independent and know only their own status, they need to exchange signaling packets, which also waste link bandwidth and device resources.





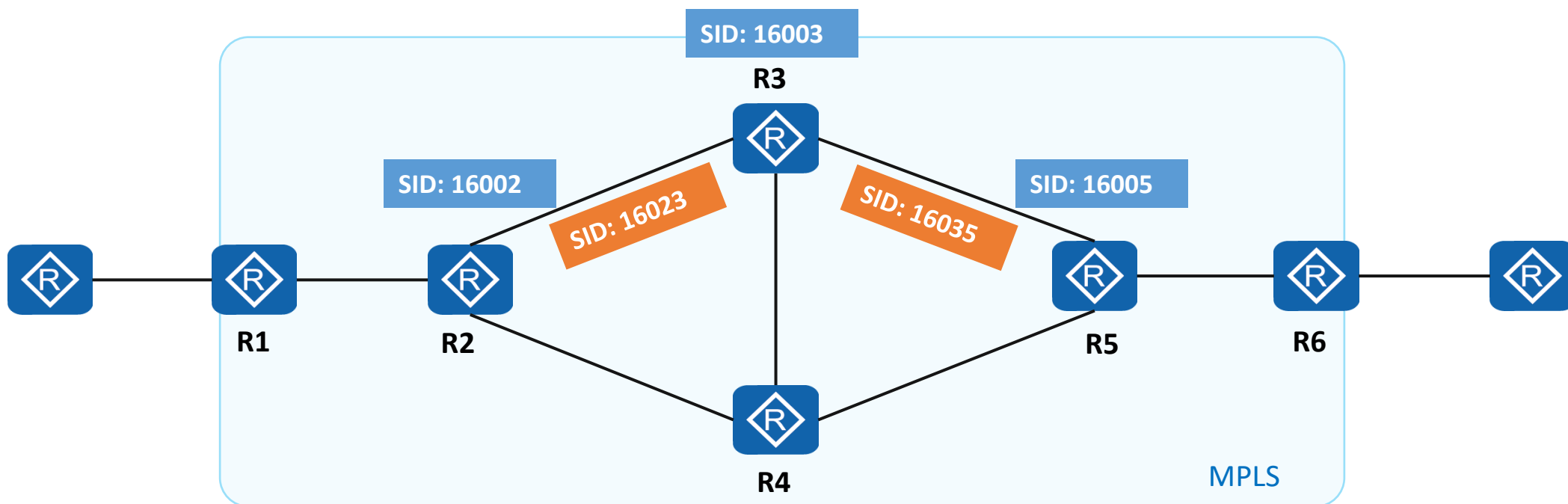
Introduction to Segment Routing

- To solve the problems facing traditional IP forwarding and MPLS forwarding, the industry proposed Segment Routing (SR). SR makes the following improvements:
 1. Extends the existing protocols.
 - The extended IGPs and BGP have the label distribution capability, eliminating the need for other label distribution protocols on networks, and thereby simplifying protocols.
 2. Introduces the source routing mechanism.
 - Using the source routing mechanism, controllers can centrally calculate paths.
 3. Allows networks to be defined by services.
 - Networks are driven by services. After service requirements, such as latency, bandwidth, and packet loss rate requirements, are raised by applications, a controller can collect information such as the network topology, bandwidth usage, and latency, and calculate explicit paths based on these requirements.



SR Forwarding Implementation (1)

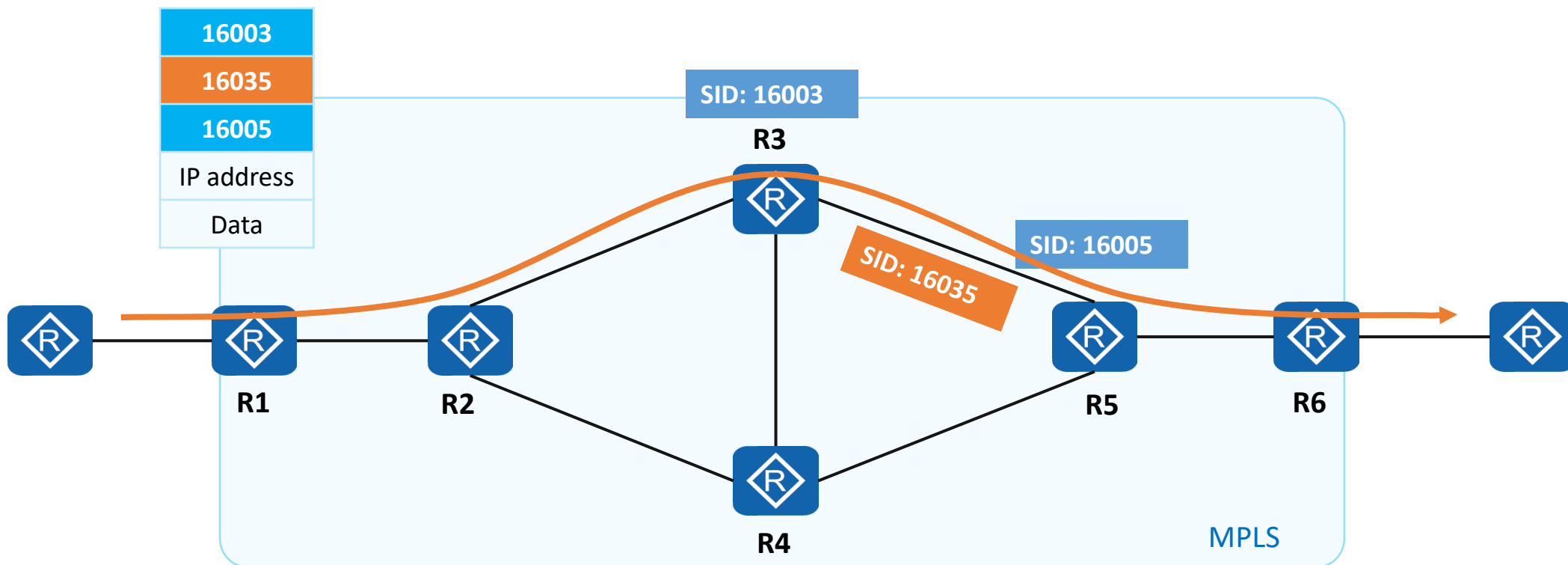
- SR divides a network path into segments and assigns segment IDs (SIDs) to these segments.
- SIDs are allocated to forwarding nodes or adjacency links. In this example, SIDs of the forwarding nodes are expressed in 1600X, where X is a node ID; SIDs of the adjacency links are expressed in 160XX, where XX indicates the node IDs at both ends of a link.





SR Forwarding Implementation (2)

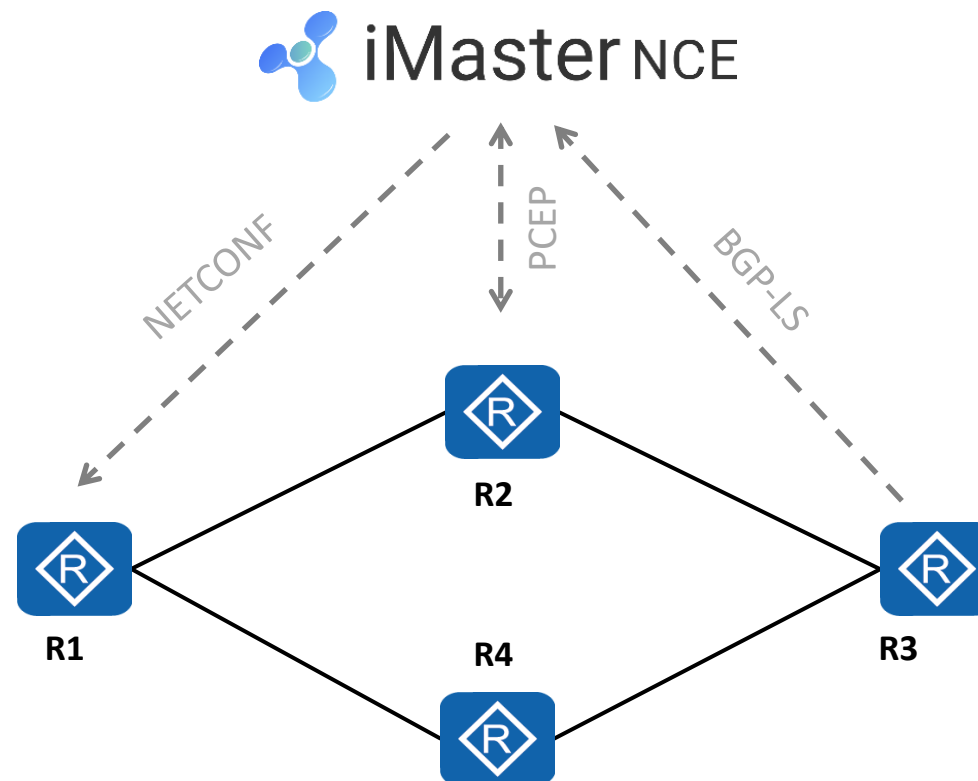
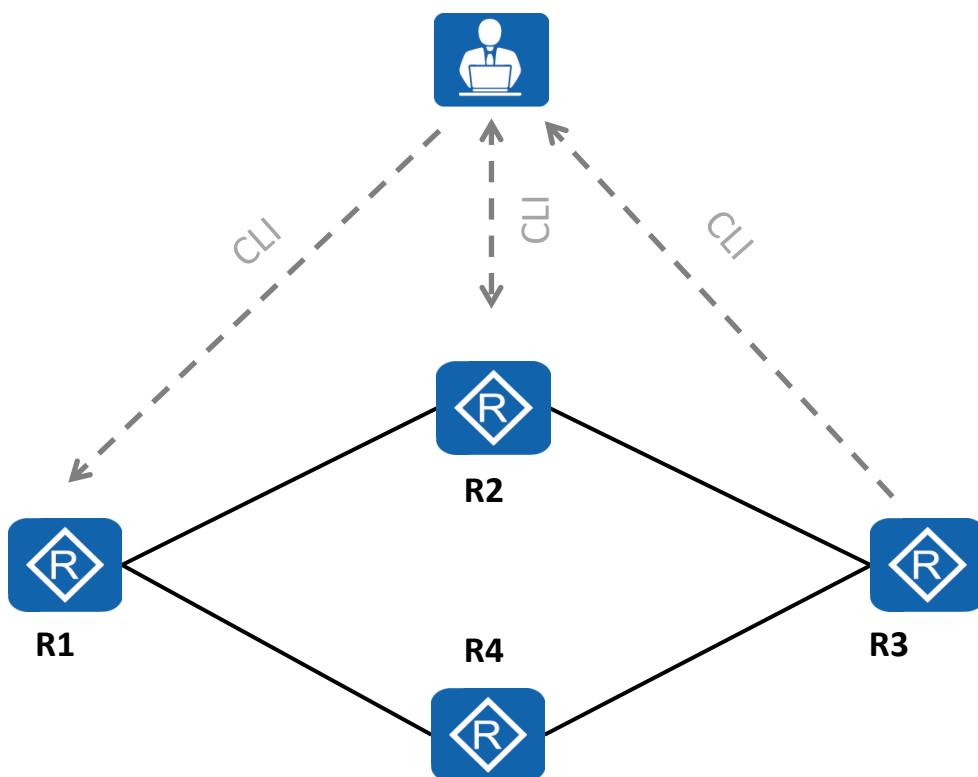
- SIDs of adjacency links and network nodes are arranged in order to form a segment list, which represents a forwarding path. The segment list is encoded by the source node in a header of a data packet, and is transmitted with the data packet. The essence of SR is instructions, which guide where and how packets go.





SR Deployment Modes

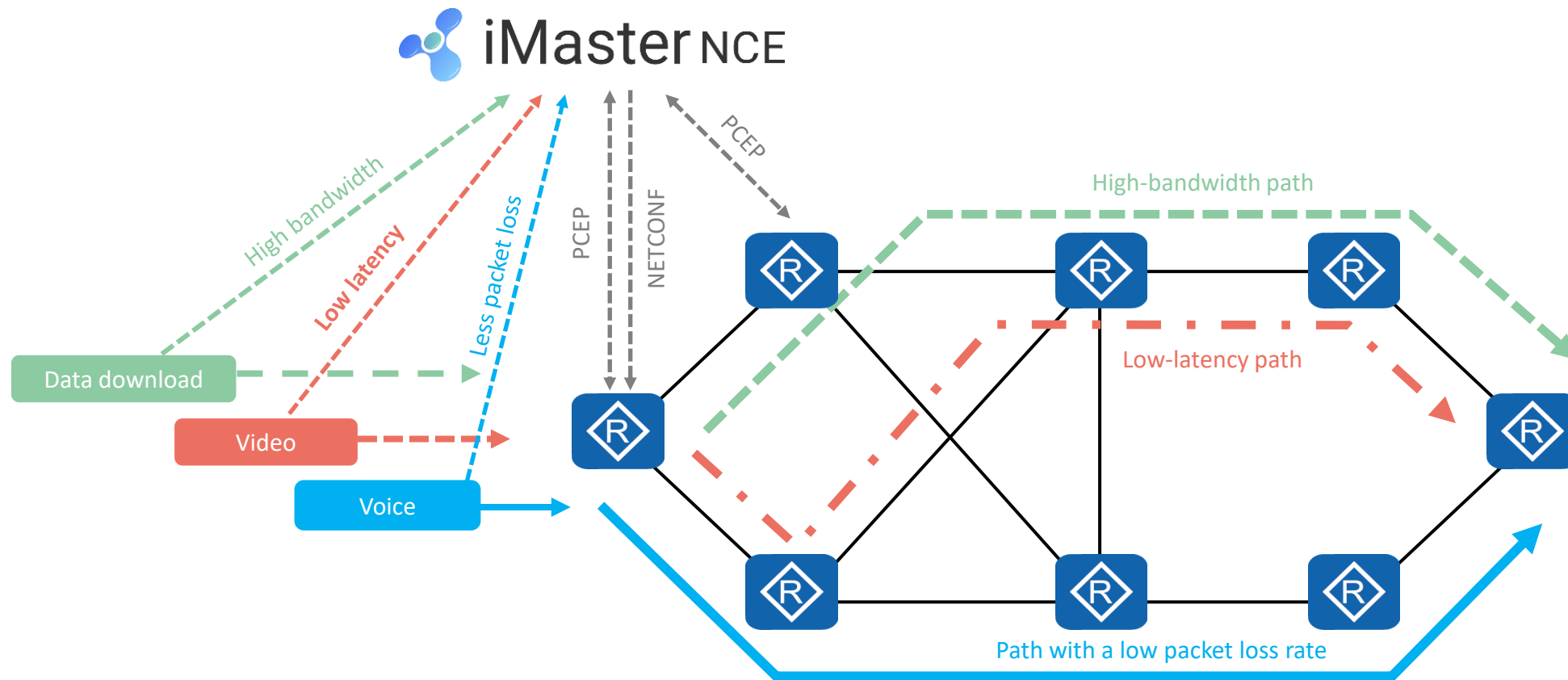
- SR can be deployed with or without a controller. If a controller is used, the controller collects information, reserves path resources, computes paths, and delivers the results to the source node. This mode is preferred.





SR Application

- SR can be used to easily specify packet forwarding paths. On a live network, different paths can be defined for different services. In this example, three explicit paths are defined to implement the service-driven network: one each for data download, video, and voice services. Devices are managed by the controller, which can quickly provision paths in real time.





Quiz

1. (Multiple) Which of the following statements about PPP are true?
 - A. PPP supports the bundling of multiple physical links into a logical link to increase the bandwidth.
 - B. PPP supports cleartext and ciphertext authentication.
 - C. PPP cannot be deployed on Ethernet links because of its poor scalability.
 - D. PPP supports asynchronous and synchronous links for the physical layer.
 - E. PPP supports multiple network layer protocols, such as IPCP.
2. (Single) After a PPPoE client sends a PADI packet to PPPoE servers, the PPPoE servers reply with a PADO packet. Which kind of frame is the PADO packet?
 - A. A. Multicast B. Broadcast C. Unicast D. Anycast
3. (Single) Which of the following values of the Length/Type field in an Ethernet data frame indicates that the Ethernet data frame carries PPPoE discovery packets?
 - A. A. 0x0800 B. 0x8864 C. 0x8863 D. 0x0806



Summary

- This course reviews the types and applications of early WAN technologies and describes the evolution of WANs from the early circuit switching networks to IP networks, MPLS label switching network, and finally to SR networks. With the development of network technologies, networks become more efficient and intelligent.
- The course also describes the implementation of PPP, including parameter negotiation during PPP link establishment, authentication negotiation, and network layer negotiation. It analyzes in detail two PPP authentication protocols – PAP and CHAP, and describes their working processes and differences.
- PPPoE is the most widely used PPP application. By analyzing how a PPPoE session is discovered, negotiated, established, and torn down, this course help you better understand the working mechanism and configuration of PPPoE.



More Information

- (Multimedia) Segment Routing MPLS Advanced Series
 - <https://support.huawei.com/carrier/docview?nid=DOC1100645168&path=PBI1-7275726/PBI1-21782273/PBI1-7275849/PBI1-7276518/PBI1-15837>
- (Multimedia) Segment Routing IPv6 Advanced Series
 - <https://support.huawei.com/enterprise/en/doc/EDOC1100133514?idPath=24030814%7C9856750%7C22715517%7C9858933%7C15837>

The image features a blue-tinted background showing silhouettes of several business professionals in a modern office environment. They are standing on a reflective floor, and their reflections are visible below them. In the background, a city skyline with tall buildings is visible through large windows. The overall atmosphere is professional and corporate.

Thank You
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